Kevin Zhou

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Employment and Education

UC Berkeley / Lawrence Berkeley National Lab 2024–present
Postdoctoral Researcher

Stanford University 2024
PhD in Physics
Advisor: Natalia Toro, Thesis: *Novel Searches for Physics Beyond the Standard Model*

Oxford University (New College) 2019
MSc in Mathematical and Theoretical Physics with distinction

Cambridge University (St. John's College) 2018
MASt in Mathematics with distinction

Massachusetts Institute of Technology 2017
BS in Physics and Mathematics

Fellowships and Awards

NSF Graduate Research Fellowship	2017 – 2022
Marshall Scholarship	2017 – 2019
Demuth Prize, New College	2019
Dirac Prize, St. John's College	2018
Finalist, Hertz Fellowship	2017
Joel Matthew Orloff Award for Outstanding Research, MIT	2017
Honorable Mention, Putnam Mathematical Competition	2016, 2017
Gold Medal, International Physics Olympiad	2012, 2013

Publications

Most of these works have alphabetical author lists; exceptions are underlined. Key papers are starred.

2609.10671* Ultralight Axial Dark Matter
A. Hook, J. Huang, K. Zhou, under review at JHEP

2607.27313* Suppressed Quantum Effects of Weakly Coupled Waves
Y. Bao, D. Y. Cheong, N. Rodd, J. Takach, L.-T. Wang, K. Zhou, under review at Phys. Rev. X

2511.07514 Quantum Calculations of the Cavity Shift in Electron Magnetic Moment Measurements
F. Day, R. Harnik, Y. Kahn, S. Pavaskar, K. Zhou, JHEP 03, 038 (2026)

2510.05198* Intrinsically Quantum Effects of Axion Dark Matter are Undetectable
Y. Bao, D. Y. Cheong, N. Rodd, J. Takach, L.-T. Wang, K. Zhou,
Phys. Rev. Lett. 136, 171601 (2026)

2507.07173 A Prototype Hybrid Mode Cavity for Heterodyne Axion Detection
Z. Li, K. Zhou, M. Oriunno, A. Berlin, S. Calatroni, R. T. D'Agnolo,
S. A. R. Ellis, P. Schuster, S. Tantawi, N. Toro, NIMA 171474 (2026)

2504.02927 Determining Spin-Dependent Light Dark Matter Rates from Neutron Scattering
A. Berlin, A. J. Millar, T. Trickle, K. Zhou, Phys. Rev. D 112, 035021 (2025)

2502.01725 Ponderomotive Effects of Ultralight Dark Matter
K. Zhou, JHEP 05, 134 (2025)

2312.11601 Physical Signatures of Fermion-Coupled Axion Dark Matter
A. Berlin, A. J. Millar, T. Trickle, K. Zhou, JHEP 05, 314 (2024)

2303.04816* Interactions of Particles with "Continuous Spin" Fields
P. Schuster, N. Toro, K. Zhou, JHEP 04, 010 (2023)

2209.12901 Discovering QCD-Coupled Axion Dark Matter with Polarization Haloscopes
A. Berlin, K. Zhou, Phys. Rev. D 108, 035038 (2023)

2112.02104 Probing Invisible Vector Meson Decays with the NA64 and LDMX Experiments
P. Schuster, N. Toro, K. Zhou, Phys. Rev. D 105, 035036 (2022)

2106.09033* Stellar Shocks From Dark Matter Asteroid Impacts
A. Das, S. A. R. Ellis, P. Schuster, K. Zhou, Phys. Rev. Lett. 128, 021101 (2022)

2007.15656 Heterodyne Broadband Detection of Axion Dark Matter
A. Berlin, R. T. D'Agnolo, S. A. R. Ellis, K. Zhou, Phys. Rev. D 104, L111701 (2021)

1912.11048* Axion Dark Matter Detection by Superconducting Resonant Frequency Conversion
A. Berlin, R. T. D'Agnolo, S. A. R. Ellis, C. Nantista, J. Neilson,
P. Schuster, S. Tantawi, N. Toro, K. Zhou, JHEP 07, 088 (2020)

1704.06266 Casimir Meets Poisson: Improved Quark/Gluon Discrimination with Counting Observables
C. Frye, A. Larkoski, J. Thaler, K. Zhou, JHEP 09, 083 (2017)

1704.05456 Generalized Fragmentation Functions for Fractal Jet Observables
B. Elder, M. Procura, J. Thaler, W. Wallewijn, K. Zhou, JHEP 06, 085 (2017)

1703.04722 Minimum Energetic Cost to Maintain a Target Nonequilibrium State
J. Horowitz, K. Zhou, J. England, Phys. Rev. E 95, 042102 (2017)

White Papers and Large Collaborations

2509.26574 Probing the Critical Point of AI Reasoning: a Frontier Physics Research Benchmark
Contributed a scaffolded and fully solved example research problem

2501.14249 Humanity's Last Exam, Nature (2026)
Contributed advanced physics problems to help benchmark AI

2203.14923 Axion Dark Matter
Contributed writing for a subsection, and editing for all sections

2203.12714 Searches for New Particles, Dark Matter, and Gravitational Waves with SRF Cavities
Contributed writing and feedback

2203.08192 Current Status and Future Prospects for the Light Dark Matter eXperiment
Contributed theoretical projections for experimental sensitivity, and figures

Pedagogical Works

Physics Olympiad Handouts
Open-access notes with 1,000 solved problems, used by students in dozens of countries

2411.08283 The surprising subtlety of electrostatic field lines
K. Zhou and T. Brauner, Am. J. Phys. 93, 234–240 (2025)

Invited Seminars

Ultralight Axial Dark Matter
- Fermilab Particle Theory Seminar 8/26

Suppressed Quantum Effects of Weakly Coupled Waves
- University of Michigan Brown Bag Seminar 11/26
- University of Utah High Energy/Astrophysics Seminar 10/26
- KAIST Seminar 7/26

The Cavity Shift of Electron $g-2$
- Seoul National University Seminar 7/26
- Korea Institute for Advanced Study (KIAS) Seminar 7/26
- Southern University of Science and Technology Seminar 12/25
- UC Davis QMAP Particle/Cosmology Seminar 11/25
- Fudan University Seminar 7/25

Quantum and Classical Effects of Axion Dark Matter
- University of Toronto THEP Seminar 4/26
- Cornell Particle Theory Seminar 4/26
- Hong Kong City University Seminar 12/25
- Hong Kong University of Science and Technology Seminar 12/25
- Chinese University of Hong Kong Seminar 12/25
- UC Berkeley "4D" Seminar 10/25
- Nanjing Normal University Seminar 7/25
- Tsung-Dao Lee Institute Seminar 7/25

New Precision Experiments for Axion Dark Matter
- Institute for Basic Science (IBS) DMAG Seminar 7/26
- Tsinghua University Seminar 7/25

How (Not) to Probe the Axion-Electron Coupling
- Peking University Seminar 7/25
- UC Davis "Xperiment" Seminar 5/25
- University of Chicago Particle Theory Seminar 4/25
- UC Berkeley "4D" Seminar 8/24

Flatiron Institute, Particle Astrophysics and Cosmology Meeting Around NYC 4/24
University of Geneva High Energy Particle Physics Seminar 3/24
HEP/Astro Results Forum 3/24
SLAC Theory Seminar 11/23

Electromagnetism and Gravity with Continuous Spin
Tsinghua University Seminar 7/25
UIUC High Energy Physics Seminar 10/24
Hunting Invisibles (HIDDeN) Virtual Institute Seminar 11/23
Caltech High Energy Physics Seminar 10/23
UC Santa Cruz SCIPP Seminar 10/23
University of Maryland EPT Seminar 9/23
ICTP HECAP Seminar 7/23
CERN BSM Forum 6/23
UC Davis QMAP Particle/Cosmology Seminar 4/23
UC Berkeley "4D" Seminar 4/23
Stanford Phenomenology Seminar 2/23
Perimeter Institute Theory Seminar 10/22

Discovering the QCD Axion with Polarization Haloscopes
Fermilab Particle Theory Seminar 4/23
TRIUMF Theory Seminar 10/22
University of Victoria Theory Seminar 10/22

Flashes in the Dark: New Searches for Axions and Macroscopic Dark Matter
Johns Hopkins Theory Seminar 9/22

Searching for Ultraheavy and Ultralight Dark Matter
SLAC Theory Seminar 3/22

Heterodyne Detection of Axion Dark Matter
Virtual Axion Institute 8/20

Invited Conference/Workshop Talks

Suppressed Quantum Effects of Weakly Coupled Waves
Bay Area Particle Theory Seminar 10/26

Ultralight Axial Dark Matter
Dark Frontiers 2026 (plenary) 8/26

Quantum and Classical Effects of Axion Dark Matter
Zhejiang University Institute of Quantum Sensing International Workshop 4/26

The Superconducting Heterodyne Approach to Axion Detection
Tsung-Dao Lee Institute Mini-Workshop on Cavities for Fundamental Physics 4/26
Berkeley Axion Workshop 2025 5/25

New Precision Experiments for Axion Dark Matter
NOPP 2025: International Workshop on New Opportunities for Particle Physics (plenary) 7/25

Contributed Conference Talks

The Superconducting Heterodyne Approach to Axion Detection
- Axions in Stockholm 2025 — 7/25
- Axion 2025: 4th International Conference on Axion Physics and Experiment — 7/25

Spin-Dependent Dark Matter Rates from Neutron Scattering
- Phenomenology 2025 Symposium — 5/25

Physical Signatures of Fermion-Coupled Axion Dark Matter
- Phenomenology 2024 Symposium — 5/24

Discovering the QCD Axion with Polarization Haloscopes
- 18th Patras Workshop on Axions, WIMPs and WISPs — 7/23
- Phenomenology 2023 Symposium — 5/23

Probing Dark Sectors With Invisible Vector Meson Decays
- Phenomenology 2022 Symposium — 5/22
- APS April Meeting 2022 — 4/22
- ILCX2021: ILC Workshop on Potential Experiments — 10/21

Stellar Shocks From Dark Asteroids
- COSMO'21: 24th International Conference on Particle Physics and Cosmology — 8/21
- DPF21: APS Division of Particles & Fields Meeting — 7/21
- Phenomenology 2021 Symposium — 5/21

Contributions to Grants

An SRF Cavity for Dark Matter Axion Detection — 2022 – 2024
- SLAC LDRD grant, with principal investigator Zenghai Li
- Participated in design discussions, writing and editing of proposal and final publication

Service

Seminar and workshop organization
- Co-organized the first Heterodyne Axion Detection workshop — 2026
- Organized theory talks for the Berkeley "Axion Works" seminar — 2025, 2026
- Organized the Berkeley/LBNL particle theory seminar — 2025

Department service
- Served as student representative for the physics department's Graduate Studies Committee — 2023
- Participated on several panels for undergraduates and incoming graduate students — 2020

Peer review
- Refereed research papers for Phys. Rev. Lett, Phys. Rev. D, Nature Communications, JHEP, JCAP, Phys. Lett. B, and Int. J. Mod. Phys. A
- Refereed pedagogical papers and books for Am. J. Phys, Cambridge University Press, and World Scientific

Outreach

Press coverage

- Interviewed in the New York Times for work on "Humanity's Last Exam" 2025
- Participated in several interviews for "Stellar Shocks From Dark Matter Asteroid Impacts" 2022
  (Altmetric score of 200+, in top 1% of Physical Review Letters)

Online discussion

- Wrote 1,000 answers on StackExchange for all fields of physics (3 million total views)
- Moderator of Reddit's r/Physics, the internet's most active physics forum (1 million visitors per month)

National Science Bowl 2023, 2024

- Wrote and edited physics questions for the U.S. Department of Energy's flagship middle school and high school outreach event (∼10,000 participants)

Local outreach

- Mentored a San Jose State University undergrad for the DOE's RENEW-HEP program Summer 2024
- Judged research presentations for the US Invitational Young Physicists Tournament 2023
- Taught 250+ high school students at "Splash" events at MIT, Oxford, and Stanford 2013 – 2019

Teaching

U.S. Physics Olympiad 2015 – 2024

- Wrote and edited the "$F = ma$ exam" (largest American physics competition, 6,000 annual participants)
- Wrote and edited the USAPhO (most challenging American physics competition, 500 annual participants)
- Taught theoretical and experimental classes for finalists at annual training camps
- Managed the theory training of America's team for the European and International Physics Olympiads
- Wrote and released training materials used in North America, Europe, and Asia

Graduate Quantum Mechanics I, Stanford (TA) 2024

Intermediate Electricity and Magnetism I, Stanford (TA) 2023

Quantum Field Theory I, Stanford (TA, overall rating 4.7/5.0) 2022

- Ran weekly sections and office hours; helped write, edit, solve, and grade new problem sets